Model Performance Measurement activity

I created separate evaluation cells using distinct model variables to avoid affecting later code.

I varied the LogisticRegression regularization strength C and observed its impact on AUC.

```
from sklearn.datasets import load_iris
from sklearn.linear_model import LogisticRegression
from sklearn.metrics import roc_auc_score

X, y = load_iris(return_X_y=True)

# Baseline
clf_base = LogisticRegression(solver="lbfgs", max_iter=2000).fit(X, y)
auc_ovr_base = roc_auc_score(y, clf_base.predict_proba(X), multi_class="ovr")
auc_ovo_base = roc_auc_score(y, clf_base.predict_proba(X), multi_class="ovo")

# Change 1: stronger regularisation (smaller C)
clf_c001 = LogisticRegression(solver="lbfgs", C=0.01, max_iter=2000).fit(X, y)
auc_ovr_c001 = roc_auc_score(y, clf_c001.predict_proba(X), multi_class="ovr")

# Change 2: weaker regularisation (bigger C)
clf_c10 = LogisticRegression(solver="lbfgs", C=10, max_iter=2000).fit(X, y)
auc_ovr_c10 = roc_auc_score(y, clf_c10.predict_proba(X), multi_class="ovr")

print("Iris AUC results")
print(f"Baseline   OVR={auc_ovr_base:.4f}  OVO={auc_ovo_base:.4f}")
print(f"C=0.01     OVR={auc_ovr_c001:.4f}")
print(f"C=10       OVR={auc_ovr_c10:.4f}")
```

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```
Iris AUC results
Baseline   OVR=0.9984  OVO=0.9984
C=0.01     OVR=0.9805
C=10       OVR=0.9989
```

Iris (multiclass, OVR AUC): baseline 0.9984, C=0.01 dropped to 0.9805, C=10 increased slightly to 0.9989.

Breast cancer (binary AUC on a train test split): baseline C=1.0 was 0.9914, C=0.01 dropped to 0.9787.

```
from sklearn.datasets import load_breast_cancer
from sklearn.model_selection import train_test_split
from sklearn.metrics import roc_auc_score
from sklearn.linear_model import LogisticRegression

Xb, yb = load_breast_cancer(return_X_y=True)
Xtr, Xte, ytr, yte = train_test_split(Xb, yb, test_size=0.3, random_state=0, stratify=yb)

clf_b_base = LogisticRegression(solver="liblinear", C=1.0, max_iter=2000).fit(Xtr, ytr)
auc_b_base = roc_auc_score(yte, clf_b_base.predict_proba(Xte)[:, 1])

clf_b_c001 = LogisticRegression(solver="liblinear", C=0.01, max_iter=2000).fit(Xtr, ytr)
auc_b_c001 = roc_auc_score(yte, clf_b_c001.predict_proba(Xte)[:, 1])

print("Breast cancer AUC results")
print(f"Baseline C=1.0   AUC={auc_b_base:.4f}")
print(f"C=0.01           AUC={auc_b_c001:.4f}")
```

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```
Breast cancer AUC results
Baseline C=1.0   AUC=0.9914
C=0.01           AUC=0.9787
```

Interpretation: stronger regularization, smaller C, reduced discriminative performance in both cases, while weaker regularization kept AUC high. This shows why model selection needs controlled metric comparisons, not a single run.